La Scuola d'Italia / Dr. Huson / IB Math 11 (a)
28 October 2026

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2.6 Quiz: Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. [Maximum mark: 10]

   Given the points $A(-4,\ 1)$, $B(4,\ 5)$ and $C(1,\ 6)$.

   (a) Find the gradient of the line AB. [2]

   (b) Find the equation of the line AB. [2]

   The line L passes through C and is perpendicular to the line AB.

   (c) Find the equation of L. [3]

   (d) Find the coordinates of the point where L meets the line AB. [3]

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2. [Maximum mark: 7]

At a cinema, an adult ticket costs x euros and a child ticket costs y euros. A group of 2 adults and 3 children pays 47 euros. A group of 1 adult and 4 children pays 41 euros.

(a) Write down two equations in x and y. [2]

(b) Find the price of an adult ticket and the price of a child ticket. [3]

(c) A family of 2 adults and some children pays 61 euros. Find the number of children in the family. [2]

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3. [Maximum mark: 10]

(a) Simplify $(2a^3b)^2 \times 3a^{-2}b^{-4}$. Give your answer using positive exponents only. [3]

(b) Find the exact value of $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$. [2]

(c) Solve the equation $9^{x-1} = 27^{x-3}$. [3]

(d) Solve the equation $2^{x+3} = \frac{1}{32}$. [2]

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4. [Maximum mark: 8]

In this question, give every answer in exact form.

(a) Simplify $\sqrt{50} + \sqrt{8} - \sqrt{32}$. [2]

(b) Expand and simplify $\left(3 + \sqrt{5}\right)\left(3 - \sqrt{5}\right)$. [2]

(c) Hence, write $\dfrac{8}{3 - \sqrt{5}}$ in the form $a + b\sqrt{5}$, where $a, b \in \mathbb{Z}$. [2]

(d) Solve the equation $x\sqrt{2} + 4 = 3x\sqrt{2}$. Give your answer in its simplest form. [2]

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2.6 Quiz: Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. [Maximum mark: 12]

   The line L_1 has equation $3x + 4y = 24$. It meets the x-axis at the point A and the y-axis at the point B. The point P has coordinates $(-1,\ -3)$.

   (a) Find the gradient of L_1. [2]

   The line L_2 passes through P and is perpendicular to L_1.

   (b) Find the equation of L_2. [3]

   (c) Find the coordinates of the point F where L_1 and L_2 intersect. [3]

   (d) Find the area of triangle PAB. [4]

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2. [Maximum mark: 7]

Solve the following system of equations by elimination. Show your working. [7]

$$2x + 3y - 4z = -20$$
$$5x - 2y + 3z = 34$$
$$3x + 4y + 2z = 11$$

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3. [Maximum mark: 11]

(a) Solve the equation $25^{x+1} = 125^{2x-3}$. [3]

(b) Find the integers a and b such that $\left(2+\sqrt{3}\right)\left(a+b\sqrt{3}\right) = 12+7\sqrt{3}$. [3]

Consider the equation $3\left(9^x\right) - 82\left(3^x\right) + 27 = 0$.

(c) By letting $u = 3^x$, solve the equation. [5]

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4. [Maximum mark: 8]

An arithmetic sequence has first term $u_1 = 7$ and common difference 3. A geometric sequence has first term $v_1 = 2$ and common ratio 1.5.

(a) Write down an expression for u_n and an expression for v_n. [2]

(b) Find the least value of n for which $v_n > u_n$. [3]

(c) Find the least value of n for which the sum of the first n terms of the geometric sequence is greater than the sum of the first n terms of the arithmetic sequence. [3]